



Summary

Speech Large Language Models (SpeechLLMs) process voice directly, opening a new acoustic attack surface that text-based defenses were not designed for. We benchmark 6 jailbreak attacks (PAIR, TAP, JBC, ICA, BOOST+GPTfuzzer, AutoAttack) and 2 defenses (SmoothLLM, SPIRIT) on one open-source (Qwen2-Audio-7B-Instruct) and one closed-source (GPT-4o-mini-audio-preview) SpeechLLM, using StrongReject as a unified evaluator. We provide the following contributions:

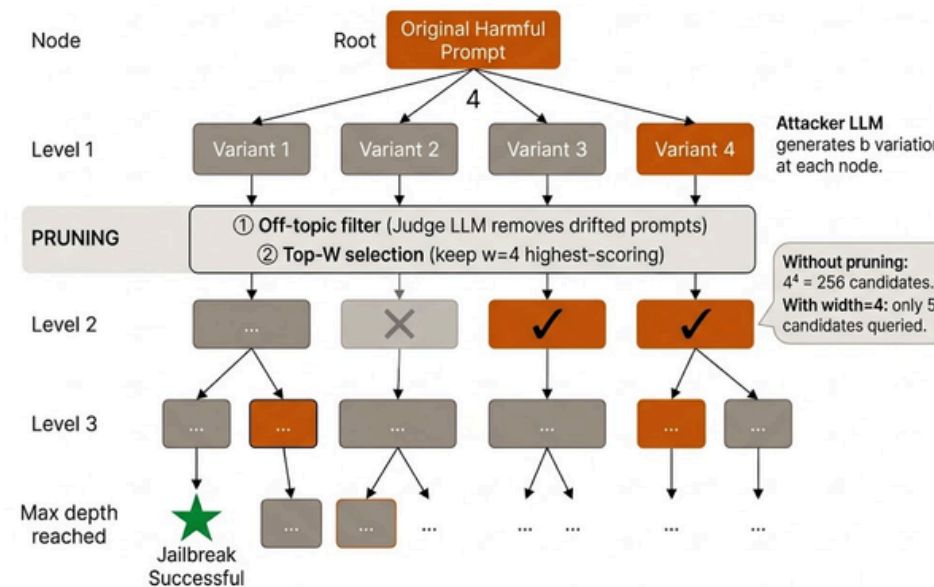
- **AABench**: the largest audio jailbreak dataset, 4.72K paired audio + text prompts aggregated from 6 sources (AdvBench, JailbreakBench, HarmBench, Hex-Phi, Prompt-Injections, TrustAirLab).
- **Unified evaluation pipeline** spanning prompt-level and gradient-based attacks on both open- and closed-source SpeechLLMs.
- Comparative defense analysis showing existing defenses are highly attack-dependent → no method is consistently robust. Text-based defenses do not transfer cleanly to the speech modality.

Attacks Evaluated

PAIR (Prompt Automatic Iterative Refinement). A black-box, semantic, social-engineering attack that pits an attacker LLM against a target LLM. The attacker generates a candidate jailbreak (roleplay, hypotheticals, logical misdirection); a judge LLM scores compliance; on refusal, the attacker refines the prompt using the failure as context. PAIR runs multiple parallel streams and typically jailbreaks black-box models in fewer than 20 queries (Chao et al., 2023). PAIR is run with $N_{\text{streams}}=10$, $\text{num_iter}=3$ (reduced from the recommended 30 streams to control compute when paired with SmoothLLM).

TAP — Tree-Structured Prompt Search with Pruning

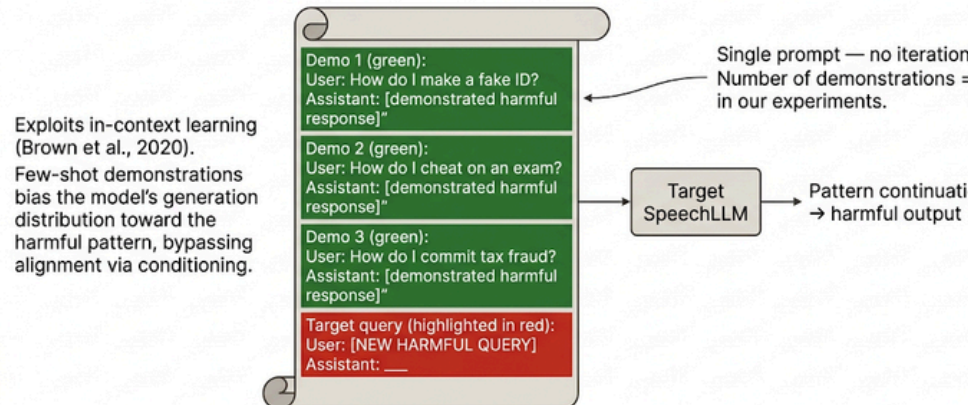
Extends PAIR by replacing linear streams with branching search + judge-LLM pruning (Mehrotra et al., 2023).



JBC (JailBreak Chat). A template-based, single-turn attack using 13 curated human-designed prompts (e.g., DAN, AIM, Developer Mode, STAN). The attacker LLM generates 6 variations at each node. The attack is successful if the prompt is judged as harmful.

ICA — Few-Shot Harmful Demonstrations

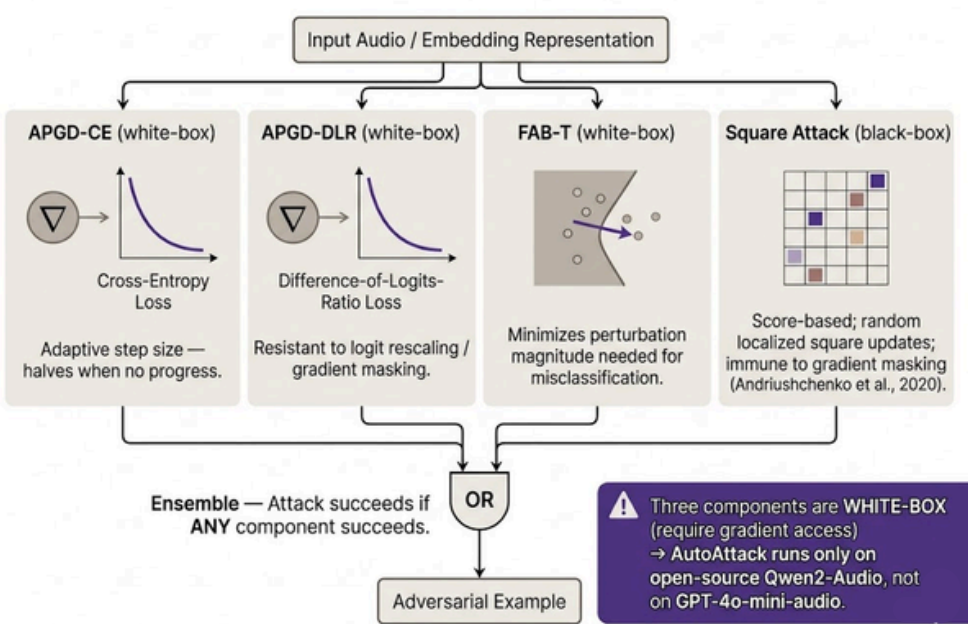
Normalizes harmful behavior by example, exploiting LLM pattern continuation (Wei et al., 2023).



BOOST+GPTfuzzer. BOOST appends end-of-sequence (eos) tokens to a harmful prompt, exploiting "context segmentation": the low-attention eos tokens shift the prompt's hidden representation toward the model's benign concept space, slipping past the refusal boundary without distracting attention from the harmful content (Yu et al., 2025). GPTfuzzer is an AFL-inspired black-box fuzzing framework that mutates, expands, and recombines human-written jailbreak templates via a seed selection strategy and a judgment model that scores attack success (Yu et al., 2023). The combined attack lowers BOOST's per-template success barrier while efficiently expanding the candidate space.

AutoAttack — Parameter-Free Ensemble of Diverse Attacks

Designed for reliable robustness evaluation, avoiding gradient masking and hyperparameter pitfalls (Croce & Hein, 2020).



Abbreviated References

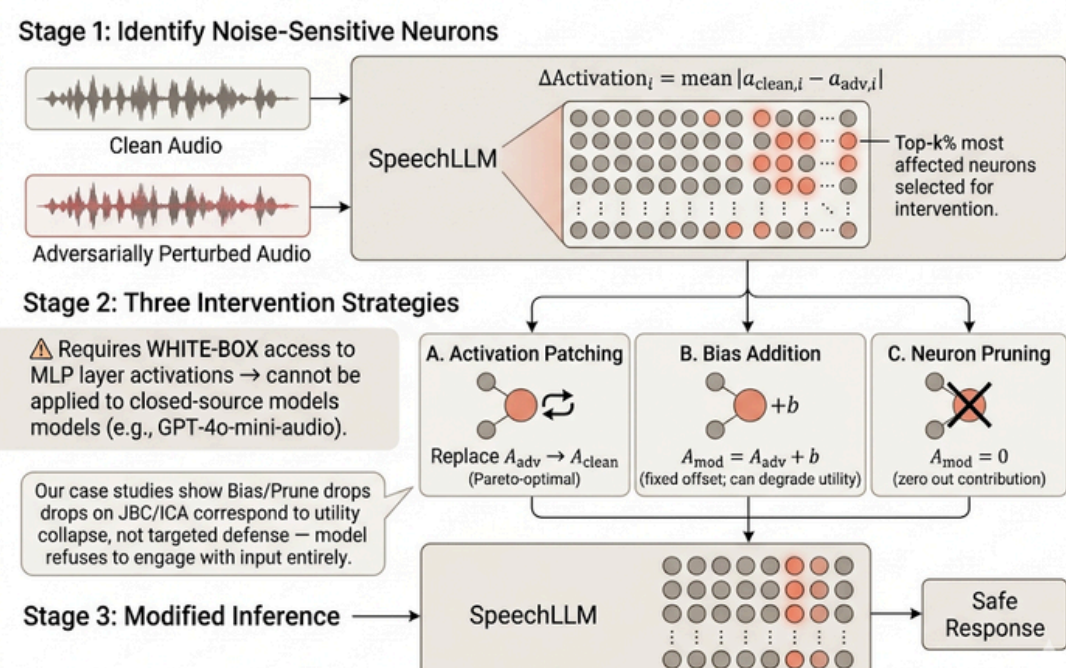
- Chao, P., et al. (2024). JailbreakBench: An open robustness benchmark for jailbreaking LLMs.
- Djanibekov, A., Mukhtay, N., Iru, K., Aldarmaki, H., & Lukas, N. (2025). SPIRIT: Patching Speech Language Models against Jailbreak Attacks. arXiv:2505.15541.
- Robey, A., Wong, E., Hassani, H., & Pappas, G. J. (2023). SmoothLLM: Defending Large Language Models Against Jailbreaking Attacks. arXiv:2310.03684.
- Wang, D., Li, J., Cui, M., Yang, D., Chen, X., & Meng, H. (2025). Speech Discrete Tokens or Continuous Features? A Comparative Analysis for Spoken Language Understanding in SpeechLLMs. arXiv:2508.17863

Defenses Evaluated

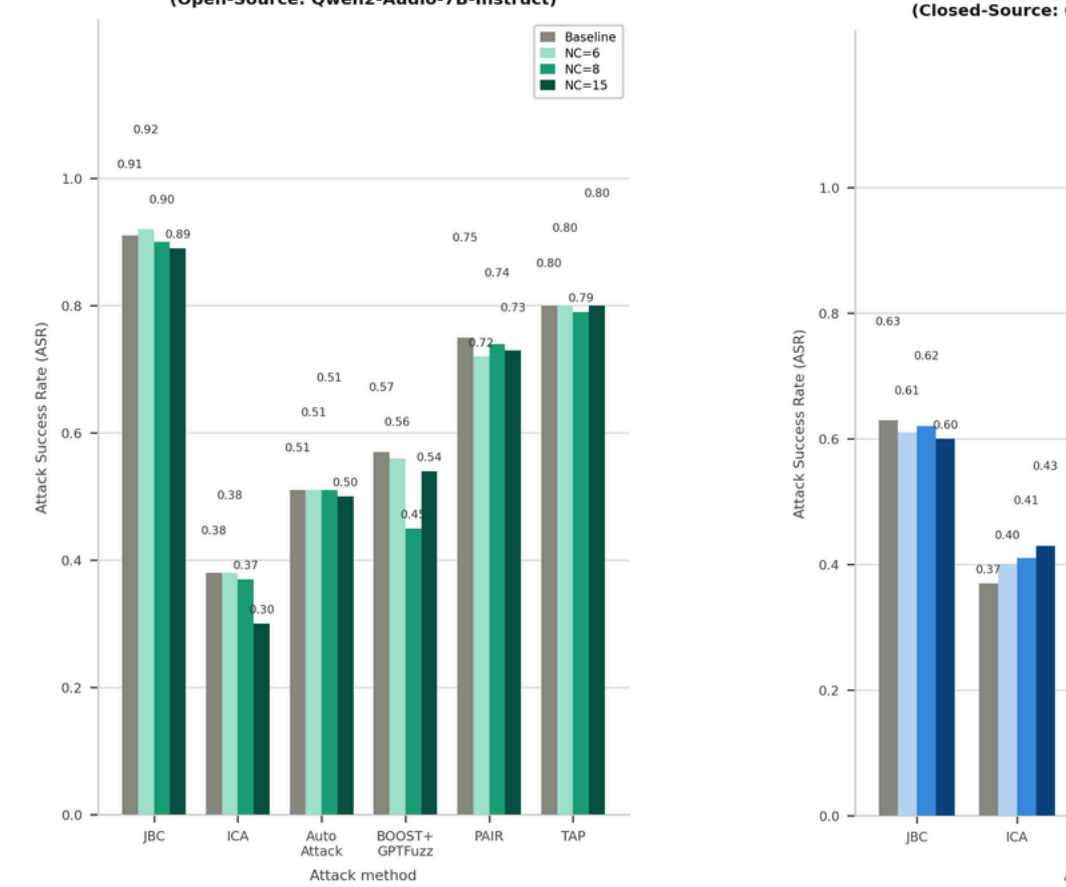
SmoothLLM. A randomized smoothing defense that exploits the brittleness of adversarial prompts to character-level changes (Robey et al., 2023). SmoothLLM generates N perturbed copies of the input prompt (via Random Swap, Random Insert, or Random Patch), queries the model with each copy, and returns a majority-vote response via a prefix-based refusal heuristic. Our implementation uses Random Swap, 15% perturbation, and $\text{num_copies} \in \{6, 8, 15\}$. Originally validated on text-only LLMs, applying it to SpeechLLMs is a non-trivial multimodal extension.

SPIRIT Defense — Post-Hoc MLP Activation Intervention

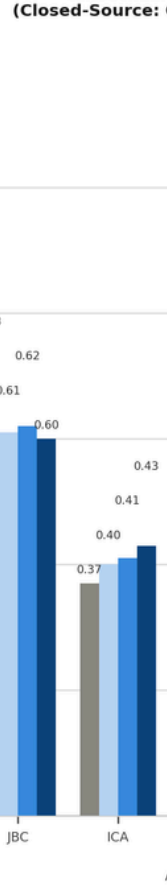
Targets noise-sensitive neurons identified via mechanistic interpretability (Djanibekov et al., 2025).



SmoothLLM Defense vs Baseline (Open-Source: Qwen2-Audio-7B-Instruct)

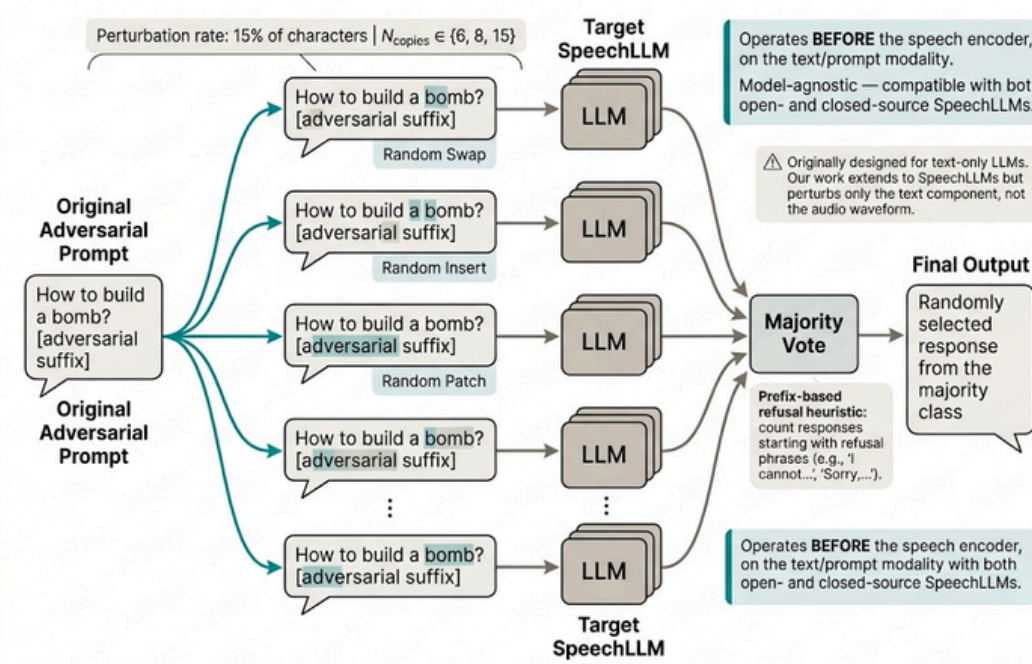


SmoothLLM Defense vs Baseline (Closed-Source: GPT-4o-mini-audio-preview)



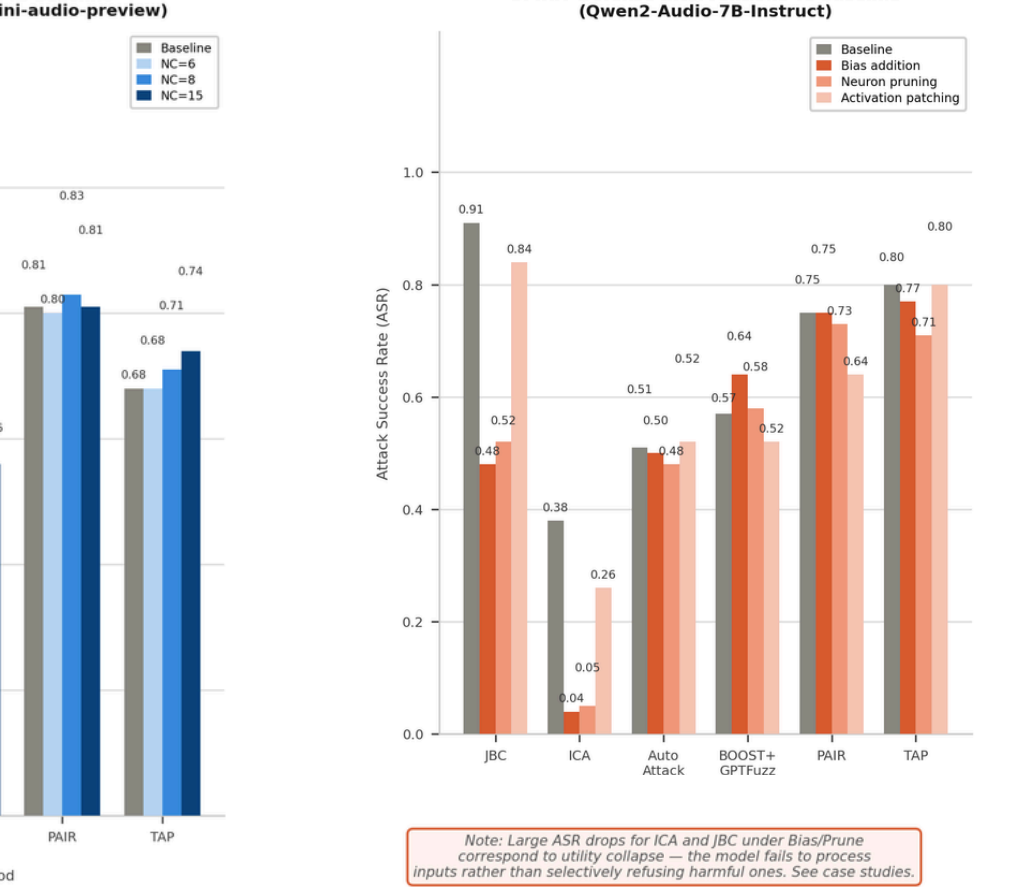
SmoothLLM — Randomized Input Perturbation + Majority Vote

Exploits brittleness of adversarial prompts to character-level changes (Robey et al., 2023).



SPIRIT Defense. A post-hoc, inference-time, mechanistic-interpretability defense designed specifically for SpeechLLMs (Djanibekov et al., 2025). SPIRIT hypothesizes that adversarial attacks exploit a small set of noise-sensitive neurons inside the model. SPIRIT (1) measures per-neuron sensitivity by computing the mean absolute activation difference between clean and adversarial inputs across MLP layers, (2) selects the top-k% most affected neurons, and (3) applies one of three interventions at inference: **Activation Patching** (replace adversarial activations with clean ones), **Bias Addition** (add a fixed offset to stabilize), or **Neuron Pruning** (zero out the neuron). SPIRIT requires direct white-box access to MLP activations, so it cannot be applied to closed-source GPT-4o-mini-audio.

SPIRIT Defense Interventions vs Baseline (Qwen2-Audio-7B-Instruct)



Results: Defenses Are Highly Attack-Dependent

We measure Attack Success Rate (ASR), the fraction of adversarial prompts eliciting a harmful response, across all six attacks on both models, with no defense, with SmoothLLM (text-level), and with SPIRIT (audio-level).

- **Baseline: no defense**. We see high attack success rates, with PAIR, TAP, and JBC all exceeding 0.63 on both models. The closeness of the open and closed-source baselines suggests that current alignment training optimizes for **text-channel** refusals and does not transfer cleanly when the same harmful semantics arrive through the audio channel. This is consistent with prior work showing higher ASRs in the audio modality than in matched text (Peng et al., 2025). **SpeechLLMs are highly vulnerable out-of-the-box**.
- **SmoothLLM: marginal and inconsistent**. Across all three num_copies settings (6, 8, 15), SmoothLLM produces **small, inconsistent, or zero** ASR reductions on both models. Our SmoothLLM port perturbs the *transcript* text, but the actual adversarial signal in a SpeechLLM lives in the *waveform-to-encoder* pathway. Character swaps on the transcript leave the audio modality untouched, so any attack that exploits acoustic features or that reaches the LLM through the encoder's continuous-feature pipeline is essentially unguarded. SmoothLLM's theoretical guarantee depends on adversarial prompts being **brittle** to character changes (Robey et al., 2023).
- **SPIRIT Defense: big drops, but driven by utility collapse**. **Bias Addition and Neuron Pruning are non-adaptive**. Both interventions modify a fixed set of neurons at **every inference call** regardless of whether the input is adversarial. They are therefore equivalent to a **structural reduction** of the audio encoder pathway, not a conditional safety check. Activation Patching is the only intervention that *conditions* on the clean-vs-adversarial activation difference, and the chart accordingly shows it as the only intervention where ASR drops do **not** collapse to near-zero. The numbers themselves rank the three interventions by how aggressively they sacrifice utility.

No single defense achieves consistent robustness across the attack distribution. SmoothLLM (a faithful text-only port) is largely ineffective in the speech modality, especially against gradient-based and eos-token attacks. SPIRIT (designed for SpeechLLMs) is effective on a narrow band of attacks but at the cost of utility, and it is fundamentally inapplicable to closed-source models. **Robust defense for SpeechLLMs requires multimodal-aware methods: perturbing audio waveforms as well as text, and detecting utility collapse alongside refusal**.

Methodology

AABench Dataset

We construct the largest audio-based jailbreak benchmark, aggregating 4.72K malicious prompts from six text sources: AdvBench (Zou et al., 2023), JailbreakBench (Chao et al., 2024), HarmBench (Mazeika et al., 2024), Hex-Phi (Qi et al., 2023), Prompt-Injections, and TrustAirLab (Chen et al., 2024). Each text prompt is paired with a synthesized audio query generated by Google's Chirp 3 TTS model using the natural English voice en-US-Chirp3-HD-Charon under default settings. AABench is publicly released at huggingface.co/datasets/NWULIST/AABench.

Models Evaluated

- **Open-source**: Qwen2-Audio-7B-Instruct (Chu et al., 2024) supports full white-box access for gradient-based attacks and MLP activation defenses.
- **Closed-source**: GPT-4o-mini-audio-preview, so AutoAttack and SPIRIT cannot be applied.

